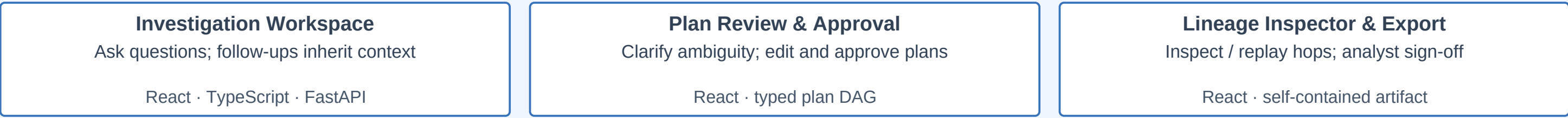


InsightPilot: System Architecture (Function + Technology)

Proposed design · Layered, agent-orchestrated architecture · CMPE 295A + 295B scope

DEPLOYMENT: ONE ORGANISATION · SELF-HOSTED / DOCKER COMPOSE

PRESENTATION TIER



question + context / analyst-approved plan

clarification / plan / answer + evidence

ORCHESTRATION TIER: CUSTOM PYTHON 3.12



typed tasks + input references + budget

results + declared grain + lineage

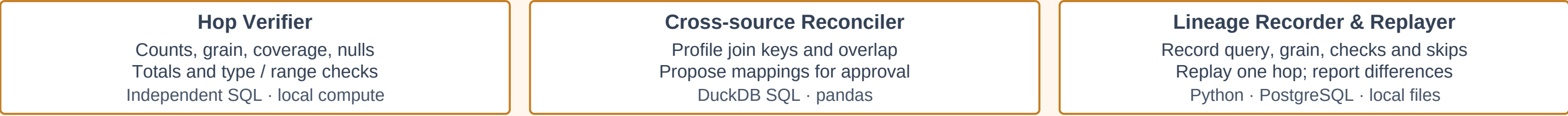
SPECIALIST AGENT TIER: ALL TASKS AND RESULTS ROUTED THROUGH ORCHESTRATOR



per-hop outputs / verification evidence

pass / warning / failure to orchestrator

VERIFICATION & LINEAGE TIER: CHECKS USE SQL / COMPUTE, NEVER LLM SELF-CRITIQUE



independent SQL / profiles over sources

schema / row counts / coverage / grain

INTEGRATION TIER: READ-ONLY SOURCES; SEMANTIC DEFINITIONS TAKE PRECEDENCE



durable bindings, state, lineage and audit

replay metadata / evaluation outcomes

DATA, AUDIT & EVALUATION TIER



HTTPS

EXTERNAL MODEL PROVIDER

Frontier: planning / SQL
Small model: prose / chart choice
Organisation's own API key
Question + schema; optional samples
PII-filtered / schema-only mode

Outside the deployment

Returned plan / SQL / prose
Data-source access stays local

KEY: Solid box = 295A scope; dashed box + [295B] = 295B scope. Solid arrow = requests/data; dashed arrow = control/response.

AGENT CONTROL: No direct agent-to-agent calls. Failed checks halt downstream work; 295B adds capped replanning / partial-result exit.

READING: Arrows summarise tier interactions, not execution order. Analyst sign-off precedes export. DAG = directed acyclic graph; PII = personal information.